

Master Mix — bench sheet		
Ligant Bench Tools C5, engine 0.1.0. Ligant Bench Tools are free and open source. Research use only, not qualified for GxP decision-making.		
Transfer basis		preserve amount per test
Dispensed per test		50 µL
Present with cells (residual)		50 µL
Assay cells per test		1000000 cells
Samples		96
Overage		additional tests, 4
Diluent		PBS, 2% FBS
Minimum transfer		2 µL (entered)
Vessel capacity		none declared
Pipetting order		
Diluent first, then components in the order entered.		
Step	Add	Volume (µL)
1	Diluent	4400
2	Component 1, "Ab01"	10.0
3	Component 2, "Ab02"	10.0
4	Component 3, "Ab03"	10.0
5	Component 4, "Ab04"	10.0
6	Component 5, "Ab05"	10.0
7	Component 6, "Ab06"	10.0
8	Component 7, "Ab07"	10.0
9	Component 8, "Ab08"	10.0
10	Component 9, "Ab09"	10.0
11	Component 10, "Ab10"	10.0
12	Component 11, "Ab11"	10.0
13	Component 12, "Ab12"	10.0
14	Component 13, "Ab13"	10.0
15	Component 14, "Ab14"	10.0
16	Component 15, "Ab15"	10.0
17	Component 16, "Ab16"	10.0
18	Component 17, "Ab17"	10.0
19	Component 18, "Ab18"	10.0
20	Component 19, "Ab19"	10.0
21	Component 20, "Ab20"	10.0

Step	Add	Volume (μL)
22	Component 21, "Ab21"	10.0
23	Component 22, "Ab22"	10.0
24	Component 23, "Ab23"	10.0
25	Component 24, "Ab24"	10.0
26	Component 25, "Ab25"	10.0
27	Component 26, "Ab26"	10.0
28	Component 27, "Ab27"	10.0
29	Component 28, "Ab28"	10.0
30	Component 29, "Ab29"	10.0
31	Component 30, "Ab30"	10.0
32	Component 31, "Ab31"	10.0
33	Component 32, "Ab32"	10.0
34	Component 33, "Ab33"	10.0
35	Component 34, "Ab34"	10.0
36	Component 35, "Ab35"	10.0
37	Component 36, "Ab36"	10.0
38	Component 37, "Ab37"	10.0
39	Component 38, "Ab38"	10.0
40	Component 39, "Ab39"	10.0
41	Component 40, "Ab40"	10.0
42	Component 41, "Ab41"	10.0
43	Component 42, "Ab42"	10.0
44	Component 43, "Ab43"	10.0
45	Component 44, "Ab44"	10.0
46	Component 45, "Ab45"	10.0
47	Component 46, "Ab46"	10.0
48	Component 47, "Ab47"	10.0
49	Component 48, "Ab48"	10.0
50	Component 49, "Ab49"	10.0
51	Component 50, "Ab50"	10.0
52	Component 51, "Ab51"	10.0
53	Component 52, "Ab52"	10.0
54	Component 53, "Ab53"	10.0
55	Component 54, "Ab54"	10.0

Step	Add	Volume (μL)
56	Component 55, "Ab55"	10.0
57	Component 56, "Ab56"	10.0
58	Component 57, "Ab57"	10.0
59	Component 58, "Ab58"	10.0
60	Component 59, "Ab59"	10.0
61	Component 60, "Ab60"	10.0
	Total (the sum of the volumes above)	5000.0

Flags

C5-FL-01 — Established at a different staining volume. The named components were established at a different staining volume. For each: the volume it was established at, the assay staining volume, and the concentration it will be delivered at relative to the concentration it was established at, under the declared basis. Each basis is exact only in a limiting regime; in the intermediate regime neither transfers the titration exactly, and this tool cannot determine which regime applies.

- Component 7, "Ab07": established at 50 μL; assay staining volume 100 μL; under preserve amount per test, concentration in the assay ÷ concentration it was established at = 0.500.
- Component 14, "Ab14": established at 50 μL; assay staining volume 100 μL; under preserve amount per test, concentration in the assay ÷ concentration it was established at = 0.500.
- Component 21, "Ab21": established at 50 μL; assay staining volume 100 μL; under preserve amount per test, concentration in the assay ÷ concentration it was established at = 0.500.
- Component 28, "Ab28": established at 50 μL; assay staining volume 100 μL; under preserve amount per test, concentration in the assay ÷ concentration it was established at = 0.500.
- Component 35, "Ab35": established at 50 μL; assay staining volume 100 μL; under preserve amount per test, concentration in the assay ÷ concentration it was established at = 0.500.
- Component 42, "Ab42": established at 50 μL; assay staining volume 100 μL; under preserve amount per test, concentration in the assay ÷ concentration it was established at = 0.500.
- Component 49, "Ab49": established at 50 μL; assay staining volume 100 μL; under preserve amount per test, concentration in the assay ÷ concentration it was established at = 0.500.
- Component 56, "Ab56": established at 50 μL; assay staining volume 100 μL; under preserve amount per test, concentration in the assay ÷ concentration it was established at = 0.500.

C5-FL-02 — Established staining volume not recorded; ratio withheld. The named components' established staining volume is not recorded. Each is carried at its entered per-test quantity; its concentration ratio is withheld, because the concentration it was established at is unknown. The transfer basis has not been applied to it.

- Component 11, "Ab11": named.
- Component 22, "Ab22": named.
- Component 33, "Ab33": named.
- Component 44, "Ab44": named.
- Component 55, "Ab55": named.

C5-FL-04 — Not titrated in this laboratory. The named components' intended quantity was not established by titration in this laboratory. A vendor recommendation is a concentration chosen for a stated assay and is not a titrated value for this panel.

- Component 3, "Ab03": vendor recommendation.

- Component 6, "Ab06": vendor recommendation.
- Component 9, "Ab09": vendor recommendation.
- Component 12, "Ab12": vendor recommendation.
- Component 15, "Ab15": vendor recommendation.
- Component 18, "Ab18": vendor recommendation.
- Component 21, "Ab21": vendor recommendation.
- Component 24, "Ab24": vendor recommendation.
- Component 27, "Ab27": vendor recommendation.
- Component 30, "Ab30": vendor recommendation.
- Component 33, "Ab33": vendor recommendation.
- Component 36, "Ab36": vendor recommendation.
- Component 39, "Ab39": vendor recommendation.
- Component 42, "Ab42": vendor recommendation.
- Component 45, "Ab45": vendor recommendation.
- Component 48, "Ab48": vendor recommendation.
- Component 51, "Ab51": vendor recommendation.
- Component 54, "Ab54": vendor recommendation.
- Component 57, "Ab57": vendor recommendation.
- Component 60, "Ab60": vendor recommendation.

C5-FL-08 — Established at more than one staining volume. The panel's components were established at more than one staining volume; no single cocktail can deliver all of them at both their established concentration and their established amount per test. The distinct volumes and the components at each are named. Components whose established volume is not recorded are excluded from this evaluation and are named as unevaluated.

Established volumes — 50 µL: Component 7, "Ab07", Component 14, "Ab14", Component 21, "Ab21", Component 28, "Ab28", Component 35, "Ab35", Component 42, "Ab42", Component 49, "Ab49", Component 56, "Ab56"; 100 µL: Component 1, "Ab01", Component 2, "Ab02", Component 3, "Ab03", Component 4, "Ab04", Component 5, "Ab05", Component 6, "Ab06", Component 8, "Ab08", Component 9, "Ab09", Component 10, "Ab10", Component 12, "Ab12", Component 13, "Ab13", Component 15, "Ab15", Component 16, "Ab16", Component 17, "Ab17", Component 18, "Ab18", Component 19, "Ab19", Component 20, "Ab20", Component 23, "Ab23", Component 24, "Ab24", Component 25, "Ab25", Component 26, "Ab26", Component 27, "Ab27", Component 29, "Ab29", Component 30, "Ab30", Component 31, "Ab31", Component 32, "Ab32", Component 34, "Ab34", Component 36, "Ab36", Component 37, "Ab37", Component 38, "Ab38", Component 39, "Ab39", Component 40, "Ab40", Component 41, "Ab41", Component 43, "Ab43", Component 45, "Ab45", Component 46, "Ab46", Component 47, "Ab47", Component 48, "Ab48", Component 50, "Ab50", Component 51, "Ab51", Component 52, "Ab52", Component 53, "Ab53", Component 54, "Ab54", Component 57, "Ab57", Component 58, "Ab58", Component 59, "Ab59", Component 60, "Ab60". Unevaluated (established volume not recorded): Component 11, "Ab11"; Component 22, "Ab22"; Component 33, "Ab33"; Component 44, "Ab44"; Component 55, "Ab55".

- Component 1, "Ab01": named.
- Component 2, "Ab02": named.
- Component 3, "Ab03": named.
- Component 4, "Ab04": named.
- Component 5, "Ab05": named.
- Component 6, "Ab06": named.
- Component 7, "Ab07": named.
- Component 8, "Ab08": named.
- Component 9, "Ab09": named.
- Component 10, "Ab10": named.
- Component 12, "Ab12": named.

- Component 13, "Ab13": named.
- Component 14, "Ab14": named.
- Component 15, "Ab15": named.
- Component 16, "Ab16": named.
- Component 17, "Ab17": named.
- Component 18, "Ab18": named.
- Component 19, "Ab19": named.
- Component 20, "Ab20": named.
- Component 21, "Ab21": named.
- Component 23, "Ab23": named.
- Component 24, "Ab24": named.
- Component 25, "Ab25": named.
- Component 26, "Ab26": named.
- Component 27, "Ab27": named.
- Component 28, "Ab28": named.
- Component 29, "Ab29": named.
- Component 30, "Ab30": named.
- Component 31, "Ab31": named.
- Component 32, "Ab32": named.
- Component 34, "Ab34": named.
- Component 35, "Ab35": named.
- Component 36, "Ab36": named.
- Component 37, "Ab37": named.
- Component 38, "Ab38": named.
- Component 39, "Ab39": named.
- Component 40, "Ab40": named.
- Component 41, "Ab41": named.
- Component 42, "Ab42": named.
- Component 43, "Ab43": named.
- Component 45, "Ab45": named.
- Component 46, "Ab46": named.
- Component 47, "Ab47": named.
- Component 48, "Ab48": named.
- Component 49, "Ab49": named.
- Component 50, "Ab50": named.
- Component 51, "Ab51": named.
- Component 52, "Ab52": named.
- Component 53, "Ab53": named.
- Component 54, "Ab54": named.
- Component 56, "Ab56": named.
- Component 57, "Ab57": named.
- Component 58, "Ab58": named.
- Component 59, "Ab59": named.
- Component 60, "Ab60": named.

C5-FL-12 — Established cell number not recorded. The named components' established cell number is not recorded, so whether the assay reproduces the condition they were established under cannot be determined.

- Component 13, "Ab13": named.
- Component 26, "Ab26": named.

- Component 39, "Ab39": named.
- Component 52, "Ab52": named.

Visuals

C5-VZ-01 · Tube composition

present with cells 50.0 µL	dispensed 50.0 µL	assay staining volume 100 µL
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C5-VZ-02 · Cocktail composition



Total component volume ÷ total cocktail volume (antibody fraction) = 0.120. No threshold applies.

- 1 Ab01: 10.0 µL FL-08
- 2 Ab02: 10.0 µL FL-08
- 3 Ab03: 10.0 µL FL-04 FL-08
- 4 Ab04: 10.0 µL FL-08
- 5 Ab05: 10.0 µL FL-08
- 6 Ab06: 10.0 µL FL-04 FL-08
- 7 Ab07: 10.0 µL FL-01 FL-08
- 8 Ab08: 10.0 µL FL-08
- 9 Ab09: 10.0 µL FL-04 FL-08
- 10 Ab10: 10.0 µL FL-08
- 11 Ab11: 10.0 µL FL-02
- 12 Ab12: 10.0 µL FL-04 FL-08
- 13 Ab13: 10.0 µL FL-08 FL-12
- 14 Ab14: 10.0 µL FL-01 FL-08
- 15 Ab15: 10.0 µL FL-04 FL-08
- 16 Ab16: 10.0 µL FL-08
- 17 Ab17: 10.0 µL FL-08
- 18 Ab18: 10.0 µL FL-04 FL-08
- 19 Ab19: 10.0 µL FL-08
- 20 Ab20: 10.0 µL FL-08
- 21 Ab21: 10.0 µL FL-01 FL-04 FL-08
- 22 Ab22: 10.0 µL FL-02
- 23 Ab23: 10.0 µL FL-08
- 24 Ab24: 10.0 µL FL-04 FL-08
- 25 Ab25: 10.0 µL FL-08
- 26 Ab26: 10.0 µL FL-08 FL-12
- 27 Ab27: 10.0 µL FL-04 FL-08
- 28 Ab28: 10.0 µL FL-01 FL-08
- 29 Ab29: 10.0 µL FL-08
- 30 Ab30: 10.0 µL FL-04 FL-08
- 31 Ab31: 10.0 µL FL-08
- 32 Ab32: 10.0 µL FL-08
- 33 Ab33: 10.0 µL FL-02 FL-04
- 34 Ab34: 10.0 µL FL-08
- 35 Ab35: 10.0 µL FL-01 FL-08
- 36 Ab36: 10.0 µL FL-04 FL-08
- 37 Ab37: 10.0 µL FL-08
- 38 Ab38: 10.0 µL FL-08

- 39 Ab39: 10.0 µL FL-04 FL-08 FL-12
- 40 Ab40: 10.0 µL FL-08
- 41 Ab41: 10.0 µL FL-08
- 42 Ab42: 10.0 µL FL-01 FL-04 FL-08
- 43 Ab43: 10.0 µL FL-08
- 44 Ab44: 10.0 µL FL-02
- 45 Ab45: 10.0 µL FL-04 FL-08
- 46 Ab46: 10.0 µL FL-08
- 47 Ab47: 10.0 µL FL-08
- 48 Ab48: 10.0 µL FL-04 FL-08
- 49 Ab49: 10.0 µL FL-01 FL-08
- 50 Ab50: 10.0 µL FL-08
- 51 Ab51: 10.0 µL FL-04 FL-08
- 52 Ab52: 10.0 µL FL-08 FL-12
- 53 Ab53: 10.0 µL FL-08
- 54 Ab54: 10.0 µL FL-04 FL-08
- 55 Ab55: 10.0 µL FL-02
- 56 Ab56: 10.0 µL FL-01 FL-08
- 57 Ab57: 10.0 µL FL-04 FL-08
- 58 Ab58: 10.0 µL FL-08
- 59 Ab59: 10.0 µL FL-08
- 60 Ab60: 10.0 µL FL-04 FL-08
- Diluent: 4400 µL

C5-VZ-03 · Concentration ratio strip

concentration in the assay ÷ concentration established at	1/2	1	2	4
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1 Ab01 FL-08		● 1.00
2 Ab02 FL-08		● 1.00
3 Ab03 FL-04 FL-08		● 1.00
4 Ab04 FL-08		● 1.00
5 Ab05 FL-08		● 1.00
6 Ab06 FL-04 FL-08		● 1.00
7 Ab07 FL-01 FL-08	● 0.500	
8 Ab08 FL-08		● 1.00
9 Ab09 FL-04 FL-08		● 1.00
10 Ab10 FL-08		● 1.00
11 Ab11 FL-02	withheld: established staining volume not recorded (C5-FL-02)	
12 Ab12 FL-04 FL-08		● 1.00
13 Ab13 FL-08 FL-12		● 1.00
14 Ab14 FL-01 FL-08	● 0.500	
15 Ab15 FL-04 FL-08		● 1.00
16 Ab16 FL-08		● 1.00
17 Ab17 FL-08		● 1.00
18 Ab18 FL-04 FL-08		● 1.00
19 Ab19 FL-08		● 1.00
20 Ab20 FL-08		● 1.00
21 Ab21 FL-01 FL-04 FL-08	● 0.500	
22 Ab22 FL-02	withheld: established staining volume not recorded (C5-FL-02)	
23 Ab23 FL-08		● 1.00
24 Ab24 FL-04 FL-08		● 1.00
25 Ab25 FL-08		● 1.00
26 Ab26 FL-08 FL-12		● 1.00
27 Ab27 FL-04 FL-08		● 1.00
28 Ab28 FL-01 FL-08	● 0.500	
29 Ab29 FL-08		● 1.00
30 Ab30 FL-04 FL-08		● 1.00
31 Ab31 FL-08		● 1.00
32 Ab32 FL-08		● 1.00
33 Ab33 FL-02 FL-04	withheld: established staining volume not recorded (C5-FL-02)	
34 Ab34 FL-08		● 1.00
35 Ab35 FL-01 FL-08	● 0.500	
36 Ab36 FL-04 FL-08		● 1.00
37 Ab37 FL-08		● 1.00
38 Ab38 FL-08		● 1.00
39 Ab39 FL-04 FL-08 FL-12		● 1.00
40 Ab40 FL-08		● 1.00
41 Ab41 FL-08		● 1.00
42 Ab42 FL-01 FL-04 FL-08	● 0.500	
43 Ab43 FL-08		● 1.00
44 Ab44 FL-02	withheld: established staining volume not recorded (C5-FL-02)	
45 Ab45 FL-04 FL-08		● 1.00
46 Ab46 FL-08		● 1.00
47 Ab47 FL-08		● 1.00
48 Ab48 FL-04 FL-08		● 1.00
49 Ab49 FL-01 FL-08	● 0.500	
50 Ab50 FL-08		● 1.00
51 Ab51 FL-04 FL-08		● 1.00
52 Ab52 FL-08 FL-12		● 1.00
53 Ab53 FL-08		● 1.00
54 Ab54 FL-04 FL-08		● 1.00
55 Ab55 FL-02	withheld: established staining volume not recorded (C5-FL-02)	
56 Ab56 FL-01 FL-08	● 0.500	
57 Ab57 FL-04 FL-08		● 1.00
58 Ab58 FL-08		● 1.00
59 Ab59 FL-08		● 1.00
60 Ab60 FL-04 FL-08		● 1.00

This tool determines a composition. It does not verify what was prepared, or what volume reached each sample. Volumes are shown to 3 significant figures and concentrations to 6; ratios and fractions to 3 (provisional). Each number shown is rounded once, half away from zero, from the exact binary value of its own unrounded value. Each pipetted volume is rounded from its own value, and the total shown is the exact sum of the pipetted volumes as shown, so it may carry more figures. The unrounded values are in the structured result.